PCA Covariance Estimation and Closed-Form MVO

Why the Factor Estimate Beats the Sample Covariance, and MVO Problems with Explicit Solutions

IEOR E4576 — Data-Driven Models in Finance — Fall 2026 — Week 8

1 Setup

On each day $t = 1, \dots, T$ the returns $\mathbf{R}_t \in \mathbb{R}^{n\times 1}$ of n assets are i.i.d. Gaussian and satisfy

$$\mathbf{R}_t = \boldsymbol{\mu} + \mathbf{B}\mathbf{f}_t + \boldsymbol{\epsilon}_t$$

where $\mathbf{B} \in \mathbb{R}^{n\times k}$ holds the loadings, $\mathbf{f}_t \in \mathbb{R}^{k\times 1}$ the factor returns and $\boldsymbol{\epsilon}_t \in \mathbb{R}^{n\times 1}$ the idiosyncratic returns, with $\mathbf{f}_t$ and $\boldsymbol{\epsilon}_t$ independent. The factor covariance matrix is $\boldsymbol{\Sigma}_F = \mathrm{Cov}(\mathbf{f}_t) \in \mathbb{R}^{k\times k}$ and $\mathbf{D} = \mathrm{Cov}(\boldsymbol{\epsilon}_t)$ is diagonal. Hence

$$\boldsymbol{\Sigma} = \mathrm{Cov}(\mathbf{R}_t) = \mathbf{B}\boldsymbol{\Sigma}_F\mathbf{B}^T + \mathbf{D} \tag{1}$$

Let $\tilde{\mathbf{R}} \in \mathbb{R}^{n\times T}$ be the de-meaned data, $m = T - 1$, $q = n/T$, and $\hat{\boldsymbol{\Sigma}}_S = \tilde{\mathbf{R}}\tilde{\mathbf{R}}^T/m$ the sample covariance.

The portfolio problem. Estimators are compared on the factor-neutral minimum-variance problem

$$\min_{\mathbf{w}} \mathbf{w}^T\boldsymbol{\Sigma}\mathbf{w} \text{ subject to } \mathbf{C}\mathbf{w} = \mathbf{d} \tag{2}$$

with

$$\mathbf{C} = \begin{bmatrix} \mathbf{1}^T \\ \mathbf{B}^T \end{bmatrix} \in \mathbb{R}^{(k+1)\times n}$$

and $\mathbf{d} = \mathbf{e}_1 \in \mathbb{R}^{(k+1)\times 1}$, that is, $\mathbf{1}^T\mathbf{w} = 1$ and $\mathbf{B}^T\mathbf{w} = \mathbf{0}$. Assume $\mathbf{C}$ has full row rank $k+1$. For a positive definite $\boldsymbol{\Sigma}$ the solution is

$$\mathbf{w} = \boldsymbol{\Sigma}^{-1}\mathbf{C}^T\left(\mathbf{C}\boldsymbol{\Sigma}^{-1}\mathbf{C}^T\right)^{-1}\mathbf{d}$$

the case $\boldsymbol{\mu} = \mathbf{0}$ of (8) in Section 4. This is the minimum-variance problem a factor model is built for: every feasible $\mathbf{w}$ has $\mathbf{w}^T\boldsymbol{\Sigma}\mathbf{w} = \mathbf{w}^T\mathbf{D}\mathbf{w}$, so the portfolio carries only idiosyncratic risk.

Loss. Let $\hat{\mathbf{w}}$ solve (2) with an estimate $\hat{\boldsymbol{\Sigma}}$ in place of $\boldsymbol{\Sigma}$, and $\mathbf{w}^*$ with the true $\boldsymbol{\Sigma}$. The loss is the true variance of $\hat{\mathbf{w}}$ relative to the optimum:

$$L(\hat{\boldsymbol{\Sigma}}) = \frac{\hat{\mathbf{w}}^T\boldsymbol{\Sigma}\hat{\mathbf{w}}}{\mathbf{w}^{*T}\boldsymbol{\Sigma}\mathbf{w}^*} \geq 1 \tag{3}$$

Factor estimate. With the loadings $\mathbf{B}$ known, as in a fundamental factor model, the factor estimate is

$$\hat{\boldsymbol{\Sigma}}_{FM} = \mathbf{B}\hat{\boldsymbol{\Sigma}}_F\mathbf{B}^T + \hat{\mathbf{D}}$$

with $\hat{\mathbf{D}}$ diagonal. Every feasible $\mathbf{w}$ has $\mathbf{w}^T\hat{\boldsymbol{\Sigma}}_{FM}\mathbf{w} = \mathbf{w}^T\hat{\mathbf{D}}\mathbf{w}$, so $\hat{\mathbf{w}}$ depends only on $\hat{\mathbf{D}}$ and not on $\hat{\boldsymbol{\Sigma}}_F$.

2 The Comparison

Theorem 1 (Sample covariance)**.** *For every positive definite $\boldsymbol{\Sigma}$ and $T > n + 1$,*

$$\mathbb{E}\left[L(\hat{\boldsymbol{\Sigma}}_S)\right] = 1 + \frac{n-k-1}{T-n+k-1} \tag{4}$$

Theorem 2 (Factor estimate)**.** *Let $\mathbf{c}_i \in \mathbb{R}^{(k+1)\times 1}$ be the i-th column of $\mathbf{C}$, with entries 1 and the loadings of asset i. Assume the $\mathbf{c}_i$ are bounded, $\underline{D} \leq D_{ii} \leq \overline{D}$ for constants $0 < \underline{D} \leq \overline{D}$, and*

$$\frac{1}{n}\sum_{i=1}^{n}\frac{\mathbf{c}_i\mathbf{c}_i^T}{D_{ii}} \to \mathbf{G}$$

positive definite. Let the $\hat{D}_{ii}$ be independent with $\nu\hat{D}_{ii}/D_{ii} \sim \chi^2_\nu$ and $\nu > 4$. For fixed ν, as $n \to \infty$,

$$L(\hat{\boldsymbol{\Sigma}}_{FM}) \to \frac{\nu-2}{\nu-4} = 1 + \frac{2}{\nu-4} \text{ in probability} \tag{5}$$

The distributional assumption holds with $\nu = T - 1$ when $\hat{D}_{ii}$ is the sample variance of observed residuals, and with $\nu = T - k - 1$ when it is the residual variance of a time-series regression of asset i on observed factors.

Comparison. The excess loss of the sample covariance is $(n - k - 1)/(T - n + k - 1)$. It grows with n, is about $q/(1 - q)$, and $\hat{\mathbf{\Sigma}}_S$ is singular once $n \geq T$. The excess loss of the factor estimate tends to $2/(\nu - 4)$, which does not depend on n. At $n = 500$ and $T = 1000$ the two are 0.98 and 0.002.

The proofs show where the difference comes from. For the sample covariance, the error $\hat{\mathbf{w}} - \mathbf{w}^*$ has a random component in each of the $n - k - 1$ directions left free by the constraints, and each contributes $1/(T - n + k - 1)$ on average. For the factor estimate, to first order $\hat{w}_i = w_i^* x_i / \mathbb{E}[x]$ with $x_i = D_{ii}/\hat{D}_{ii}$: each position is rescaled by the estimation error of its own variance, and no position is created.

Theorem 2 takes $\mathbf{B}$ as known. PCA estimates $\mathbf{B}$ as well, and the error in its column space is not covered here.

3 Proofs

Proof of Theorem 1. Let $p = k + 1$. Substitute $\mathbf{x} = \mathbf{\Sigma}^{1/2}\mathbf{w}$ and $\mathbf{A} = \mathbf{\Sigma}^{-1/2}\mathbf{C}^T \in \mathbb{R}^{n \times p}$. The constraint becomes $\mathbf{A}^T\mathbf{x} = \mathbf{d}$, the true variance is $\|\mathbf{x}\|^2$, and the sample variance is $\mathbf{x}^T\mathbf{W}\mathbf{x}/m$ with

$$\mathbf{W} = \mathbf{\Sigma}^{-1/2}(m\hat{\mathbf{\Sigma}}_S)\mathbf{\Sigma}^{-1/2} \sim W_n(\mathbf{I}, m)$$

The law of $\mathbf{W}$ is invariant under $\mathbf{W} \mapsto \mathbf{Q}\mathbf{W}\mathbf{Q}^T$ for orthogonal $\mathbf{Q}$, so we may rotate coordinates so that the columns of $\mathbf{A}$ span the first p coordinates. Write $\mathbf{x} = (\mathbf{x}_1, \mathbf{x}_2)$ with $\mathbf{x}_1 \in \mathbb{R}^p$. The constraint fixes $\mathbf{x}_1 = \mathbf{x}_1^*$ and leaves $\mathbf{x}_2 \in \mathbb{R}^{n-p}$ free. The true optimum has $\mathbf{x}_2 = \mathbf{0}$ and variance $\|\mathbf{x}_1^*\|^2$. Partition $\mathbf{W}$ conformably into $\mathbf{W}_{11}$, $\mathbf{W}_{21}$ and $\mathbf{W}_{22}$. Minimizing $\mathbf{x}^T\mathbf{W}\mathbf{x}$ over $\mathbf{x}_2$ gives $\hat{\mathbf{x}}_2 = -\mathbf{W}_{22}^{-1}\mathbf{W}_{21}\mathbf{x}_1^*$, so

$$L = 1 + \frac{\mathbf{x}_1^{*T}\mathbf{W}_{12}\mathbf{W}_{22}^{-2}\mathbf{W}_{21}\mathbf{x}_1^*}{\|\mathbf{x}_1^*\|^2}$$

Write $\mathbf{W} = \mathbf{X}\mathbf{X}^T$ with $\mathbf{X} \in \mathbb{R}^{n \times m}$ standard Gaussian, first p rows $\mathbf{X}_1$ and remaining rows $\mathbf{X}_2$. Then $\mathbf{W}_{22} = \mathbf{X}_2\mathbf{X}_2^T \sim W_{n-p}(\mathbf{I}, m)$ and $\mathbf{W}_{21}\mathbf{x}_1^* = \mathbf{X}_2(\mathbf{X}_1^T\mathbf{x}_1^*)$. Since $\mathbf{X}_1^T\mathbf{x}_1^* \sim N(\mathbf{0}, \|\mathbf{x}_1^*\|^2\mathbf{I}_m)$ is independent of $\mathbf{X}_2$, given $\mathbf{X}_2$ the vector $\mathbf{W}_{21}\mathbf{x}_1^*$ is $N(\mathbf{0}, \|\mathbf{x}_1^*\|^2\mathbf{W}_{22})$. Hence

$$\mathbb{E}\left[L - 1 \mid \mathbf{X}_2\right] = \operatorname{tr}\left(\mathbf{W}_{22}^{-1}\right)$$

and the inverse Wishart mean $\mathbb{E}[\mathbf{W}_{22}^{-1}] = \mathbf{I}/(m - (n - p) - 1)$ gives $\mathbb{E}[L] = 1 + (n - p)/(m - n + p - 1)$, which is (4). □

Proof of Theorem 2. Every feasible $\mathbf{w}$ has $\mathbf{w}^T\mathbf{\Sigma}\mathbf{w} = \mathbf{w}^T\mathbf{D}\mathbf{w}$ and $\mathbf{w}^T\hat{\mathbf{\Sigma}}_{FM}\mathbf{w} = \mathbf{w}^T\hat{\mathbf{D}}\mathbf{w}$, so

$$\mathbf{w}^* = \mathbf{D}^{-1}\mathbf{C}^T\boldsymbol{\lambda}^*$$

$$\hat{\mathbf{w}} = \hat{\mathbf{D}}^{-1}\mathbf{C}^T\hat{\boldsymbol{\lambda}}$$

with $\boldsymbol{\lambda}^* = (\mathbf{C}\mathbf{D}^{-1}\mathbf{C}^T)^{-1}\mathbf{d}$ and $\hat{\boldsymbol{\lambda}} = (\mathbf{C}\hat{\mathbf{D}}^{-1}\mathbf{C}^T)^{-1}\mathbf{d}$. Let $x_i = D_{ii}/\hat{D}_{ii} = \nu/\chi^2_{\nu,i}$, which are i.i.d. with

$$\mathbb{E}[x] = \frac{\nu}{\nu - 2}$$

$$\mathbb{E}[x^2] = \frac{\nu^2}{(\nu - 2)(\nu - 4)}$$

Then $\hat{w}_i = x_i\mathbf{c}_i^T\hat{\boldsymbol{\lambda}}/D_{ii}$. Since the summands are independent with bounded weights, the law of large numbers gives

$$\frac{1}{n}\sum_{i=1}^{n} x_i \frac{\mathbf{c}_i\mathbf{c}_i^T}{D_{ii}} \to \mathbb{E}[x]\mathbf{G}$$

$$\frac{1}{n}\sum_{i=1}^{n} x_i^2 \frac{\mathbf{c}_i\mathbf{c}_i^T}{D_{ii}} \to \mathbb{E}[x^2]\mathbf{G}$$

so $n\hat{\boldsymbol{\lambda}} \to \mathbf{G}^{-1}\mathbf{d}/\mathbb{E}[x]$ and $n\boldsymbol{\lambda}^* \to \mathbf{G}^{-1}\mathbf{d}$. The variances are

$$n\hat{\mathbf{w}}^T\mathbf{D}\hat{\mathbf{w}} = (n\hat{\boldsymbol{\lambda}})^T\left(\frac{1}{n}\sum_{i=1}^{n} x_i^2 \frac{\mathbf{c}_i\mathbf{c}_i^T}{D_{ii}}\right)(n\hat{\boldsymbol{\lambda}}) \to \frac{\mathbb{E}[x^2]}{\mathbb{E}[x]^2}\mathbf{d}^T\mathbf{G}^{-1}\mathbf{d}$$

$$n\mathbf{w}^{*T}\mathbf{D}\mathbf{w}^* = n\mathbf{d}^T\left(\mathbf{C}\mathbf{D}^{-1}\mathbf{C}^T\right)^{-1}\mathbf{d} \to \mathbf{d}^T\mathbf{G}^{-1}\mathbf{d}$$

and their ratio tends to $\mathbb{E}[x^2]/\mathbb{E}[x]^2 = (\nu - 2)/(\nu - 4)$. □

Simulation. The script `pca_vs_sample_covariance.py` draws returns from (1) with $n = 500$, $k = 5$, loadings uniform on $[-1, 1]$ (plus 1 for the market factor), D_{ii} uniform on $[0.2, 1]$, and $\hat{D}_{ii}$ the sample variance of the residuals ($\nu = T - 1$). It averages the loss (3) over 30 draws.

		Sample covariance		Factor estimate	
T	q	Theorem 1	Simulated	Theorem 2	Simulated
2520	0.20	1.244	1.245	1.0008	1.0008
1000	0.50	1.980	1.964	1.0020	1.0020
600	0.83	5.750	5.607	1.0034	1.0034
100	5.00	—	—	1.0211	1.0210
20	25.00	—	—	1.1333	1.1282

At $q = 0.83$ the sample-covariance loss has a heavy right tail, so its 30-draw average is noisy. The last row differs from the limit because n is finite.

4 MVO Problems with Closed-Form Solutions

Consider

$$\max_{\mathbf{w}} \mathbf{w}^T \boldsymbol{\mu} - \lambda \mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w} \text{ subject to } \mathbf{C}\mathbf{w} = \mathbf{d} \tag{6}$$

with $\lambda > 0$, $\boldsymbol{\Sigma}$ positive definite and $\mathbf{C} \in \mathbb{R}^{p \times n}$ of full row rank. The objective is strictly concave and the constraints linear, so the first-order conditions characterize the unique solution. With multiplier $\boldsymbol{\nu} \in \mathbb{R}^{p \times 1}$ they are $\boldsymbol{\mu} - 2\lambda\boldsymbol{\Sigma}\mathbf{w} - \mathbf{C}^T\boldsymbol{\nu} = \mathbf{0}$, so

$$\mathbf{w} = \frac{1}{2\lambda} \boldsymbol{\Sigma}^{-1} \left(\boldsymbol{\mu} - \mathbf{C}^T \boldsymbol{\nu} \right)$$

Imposing $\mathbf{C}\mathbf{w} = \mathbf{d}$ determines $\boldsymbol{\nu}$:

$$\boldsymbol{\nu} = \left(\mathbf{C}\boldsymbol{\Sigma}^{-1}\mathbf{C}^T \right)^{-1} \left(\mathbf{C}\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} - 2\lambda\mathbf{d} \right) \tag{7}$$

$$\mathbf{w}^* = \frac{1}{2\lambda} \boldsymbol{\Sigma}^{-1} \left(\boldsymbol{\mu} - \mathbf{C}^T \boldsymbol{\nu} \right) \tag{8}$$

Two special cases, with $a = \mathbf{1}^T\boldsymbol{\Sigma}^{-1}\mathbf{1}$ and $b = \mathbf{1}^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$:

Budget constraint, $\mathbf{C} = \mathbf{1}^T$ and $\mathbf{d} = 1$. Then $\nu = (b - 2\lambda)/a$ and

$$\mathbf{w}^* = \frac{\boldsymbol{\Sigma}^{-1}\mathbf{1}}{a} + \frac{1}{2\lambda} \left(\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} - \frac{b}{a}\boldsymbol{\Sigma}^{-1}\mathbf{1} \right) \tag{9}$$

the minimum-variance portfolio plus a zero-cost portfolio scaled by $1/(2\lambda)$.

Dollar neutral, $\mathbf{C} = \mathbf{1}^T$ and $\mathbf{d} = 0$. Then $\nu = b/a$ and

$$\mathbf{w}^* = \frac{1}{2\lambda} \left(\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} - \frac{b}{a}\boldsymbol{\Sigma}^{-1}\mathbf{1} \right) \tag{10}$$

Adding a row $\boldsymbol{\beta}^T$ to $\mathbf{C}$ and a 0 to $\mathbf{d}$ makes the portfolio beta neutral as well; (7)–(8) still apply. Inequality constraints such as $w_i \geq 0$ destroy the closed form, because the set of binding constraints is not known in advance.